Linear programming and the simplex method



Optimization · Level 4 · Lesson 1

In this lesson you will

- Write a decision as a linear program in standard form.
- Explain why the simplex method walks from corner to corner.
- Read binding constraints and shadow prices, and solve an LP with GLOP.

Key ideas

- Convex region: an optimum sits at a corner
- Simplex: better neighbour after better neighbour
- Shadow price: the gain from one more unit
- Slack constraint: price zero

Activity: The garden on graph paper

1. Draw (8 min): Plot the three garden limits.
2. Corners (6 min): Find and score every corner.
3. Walk (4 min): Follow the simplex path.
4. Prices (7 min): Move the hours line out by one.

No graph paper? Use squared exercise-book paper, one square per bed. Check your corners against the lab “The best plan is at a corner”.

Let’s discuss

1. Why does convexity let simplex stop without checking every corner?
2. What does a zero shadow price tell a manager?
3. When would you choose an interior-point method over simplex?

Check yourself

- Explain in two sentences why an extra can of water adds nothing to the garden’s profit.